

B5 General Relativity — Model Solutions, Trinity 2021

Question 1 — The falling apple in General Relativity

Problem. An apple of unit mass falls a height h from a tree to the ground in vacuum. Treat the problem fully in GR: derive the geodesic equation, choose the metric, evaluate the connection coefficients, show the motion is purely radial, find $r(t)$ as seen by a nearby observer, and use a conserved energy to obtain $\frac{1}{2}(dr/d\tau)^2 = GM[1/R_E - 1/(R_E + h)]$.

One-sentence statement. Gravity is the curvature of spacetime sourced by stress–energy; free particles follow timelike geodesics of the metric.

(a) Geodesic equation from local inertial coordinates

In the apple’s local inertial frame (LIF), ξ^α are Riemann normal coordinates and τ is the proper time read off the apple’s worldline ($c^2 d\tau^2 = -\eta_{\alpha\beta} d\xi^\alpha d\xi^\beta$). Free fall:

$$\frac{d^2 \xi^\alpha}{d\tau^2} = 0. \quad (1)$$

Transform to a general chart $x^\mu(\xi)$:

$$\frac{d\xi^\alpha}{d\tau} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{dx^\mu}{d\tau}.$$

Differentiate again:

$$\frac{d^2 \xi^\alpha}{d\tau^2} = \frac{\partial \xi^\alpha}{\partial x^\mu} \frac{d^2 x^\mu}{d\tau^2} + \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}.$$

Set to zero and contract with $\partial x^\lambda / \partial \xi^\alpha$:

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma_{\mu\nu}^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0, \quad \Gamma_{\mu\nu}^\lambda \equiv \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\mu \partial x^\nu}.$$

Apple at rest in coordinates x^i : only $dx^0/d\tau = c dt/d\tau$ survives:

$$\boxed{\frac{d^2 x^\mu}{d\tau^2} + c^2 \Gamma_{00}^\mu \left(\frac{dt}{d\tau} \right)^2 = 0.} \quad (2)$$

Symbols: τ is proper time; x^μ are coordinates of the chosen chart; c converts coordinate time $t = x^0/c$; Γ_{00}^μ are Christoffel symbols of the metric in those coordinates.

(b) Choice of metric

The Earth is static, spherical, uncharged, vacuum exterior $\Rightarrow$ Schwarzschild solution (Birkhoff). Schwarzschild line element:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2), \quad (3)$$

with $r_s = 2GM/c^2$. Terms: t is the coordinate time of a distant inertial observer; r is the areal radius; (θ, ϕ) angular; M the Earth's mass.

Approximations for an apple near the Earth's surface:

- Weak field: $r_s/R_E \approx 1.4 \times 10^{-9} \ll 1$ at $r = R_E \approx 6.4 \times 10^6$ m.
- Slow motion: $|dr/dt|/c \ll 1$, so $d\tau \approx dt$ to lowest order in v/c and r_s/r .
- Static apple at $\tau = 0$: only Γ_{00}^μ matters in (2).
- Vacuum: $T_{\mu\nu} = 0$ outside the Earth, justifying Schwarzschild.

(c) Connection coefficients and radial motion

Schwarzschild metric is diagonal with $g_{00} = -c^2 A(r)$, $g_{rr} = 1/A(r)$, $g_{\theta\theta} = r^2$, $g_{\phi\phi} = r^2 \sin^2 \theta$, where $A(r) = 1 - r_s/r$.

Christoffel formula (diagonal metric):

$$\Gamma_{00}^\mu = -\frac{1}{2} g^{\mu\mu} \partial_\mu g_{00} \quad (\text{no sum}).$$

Compute each component. Time:

$$\Gamma_{00}^0 = -\frac{1}{2} g^{00} \partial_0 g_{00} = 0 \quad (\text{static}).$$

Radial:

$$\Gamma_{00}^r = -\frac{1}{2} g^{rr} \partial_r g_{00} = -\frac{1}{2} A(r) \partial_r [-c^2 A(r)] = \frac{c^2}{2} A(r) A'(r).$$

Substitute $A'(r) = r_s/r^2$:

$$\Gamma_{00}^r = \frac{c^2 r_s}{2r^2} \left(1 - \frac{r_s}{r}\right).$$

Angular (g_{00} is independent of θ, ϕ):

$$\Gamma_{00}^\theta = 0, \quad \Gamma_{00}^\phi = 0.$$

Insert into (2): only the $\mu = r$ component is non-zero:

$$\frac{d^2 r}{d\tau^2} = -c^2 \Gamma_{00}^r \left(\frac{dt}{d\tau}\right)^2 = -\frac{c^4 r_s}{2r^2} \left(1 - \frac{r_s}{r}\right) \left(\frac{dt}{d\tau}\right)^2. \quad (4)$$

Hence the motion is purely radial. $d^2\theta/d\tau^2 = d^2\phi/d\tau^2 = 0.$

Choice of time. Nearby observer under the tree is essentially at rest at $r = R_E$ in the same Schwarzschild coordinates; their proper time is $d\tau_{\text{obs}} = \sqrt{A(R_E)} dt \approx dt$ to one part in 10^9 . Use coordinate time t as the observer's clock.

Weak-field, slow-motion limit of (4): drop r_s/r in the bracket and set $dt/d\tau \rightarrow 1$:

$$\frac{d^2 r}{dt^2} \approx -\frac{c^2 r_s}{2r^2} = -\frac{GM}{r^2}.$$

Newtonian recovery on the surface, $r = R_E + z$, $z \ll R_E$:

$$\frac{d^2 z}{dt^2} \approx -g, \quad g \equiv GM/R_E^2.$$

Integrate with $z(0) = h$, $\dot{z}(0) = 0$:

$$\boxed{r(t) = R_E + h - \frac{1}{2}gt^2.} \quad (5)$$

Approximations: weak field ($r_s/R_E \rightarrow 0$), slow motion ($v \ll c$), small fall height ($h \ll R_E$ so g constant), observer's proper time $\approx t$.

(d) Conserved energy and impact velocity

Schwarzschild has Killing vector ∂_t ; for unit mass the conserved energy is

$$E = -g_{0\mu} \frac{dx^\mu}{d\tau} = c^2 A(r) \frac{dt}{d\tau}. \quad (6)$$

Initial condition: stem snaps at $r = R_E + h$ with $dr/d\tau = 0$ (apple instantaneously at rest in the LIF of the tree). Normalisation $g_{\mu\nu}u^\mu u^\nu = -c^2$:

$$-c^2 A(R_E + h) \left(\frac{dt}{d\tau} \right)_0^2 = -c^2.$$

Solve:

$$\left(\frac{dt}{d\tau} \right)_0 = A(R_E + h)^{-1/2}.$$

Substitute into (6):

$$E = c^2 A(R_E + h)^{1/2}.$$

Apply the normalisation at general r along the worldline:

$$-c^2 A(r) \left(\frac{dt}{d\tau} \right)^2 + A(r)^{-1} \left(\frac{dr}{d\tau} \right)^2 = -c^2.$$

Eliminate $dt/d\tau$ via (6), $c^2 A dt/d\tau = E$:

$$-\frac{E^2}{c^2 A(r)} + A(r)^{-1} \left(\frac{dr}{d\tau} \right)^2 = -c^2.$$

Multiply by $-A(r)/2$:

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 = \frac{E^2}{2c^2} - \frac{c^2 A(r)}{2}.$$

Insert $E^2 = c^4 A(R_E + h)$:

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 = \frac{c^2}{2} [A(R_E + h) - A(r)].$$

Substitute $A(r) = 1 - r_s/r$:

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 = \frac{c^2 r_s}{2} \left(\frac{1}{r} - \frac{1}{R_E + h} \right).$$

Use $r_s = 2GM/c^2$:

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 = GM \left(\frac{1}{r} - \frac{1}{R_E + h} \right).$$

Evaluate at impact, $r = R_E$:

$$\boxed{\frac{1}{2} \left(\frac{dr}{d\tau} \right)_{R_E}^2 = GM \left(\frac{1}{R_E} - \frac{1}{R_E + h} \right).} \quad (7)$$

Newtonian limit $h \ll R_E$: $\frac{1}{2}v^2 \approx GMh/R_E^2 = gh$. $\boxed{v = \sqrt{2gh}.}$

Question 2 — Gravitational wave emission and polarisation

Problem. (a) Identify which of four mass distributions emit GWs. (b) Compute $h_{xx}, h_{yy}, h_{xy}, h_{yx}$ from a rigid dumbbell rotating in the xy -plane, evaluated at large z . (c) Find the displacements of four free test masses, sketch over a cycle, identify polarisation. (d) Discuss polarisation, interference, refraction, diffraction for GR vs EM.

(a) Emission criteria

Linearised GR: leading radiation is mass-quadrupole. A source emits classical GWs iff its mass quadrupole moment $I_{ij} = \int \rho(x_i x_j - \frac{1}{3} \delta_{ij} r^2) d^3x$ has a non-zero *third* time derivative.

(i) **Motionless point particle.** I_{ij} constant. $\boxed{\text{No emission.}}$

(ii) **Rotating sphere (axisymmetric, constant density).** Quadrupole is invariant under the rotation. $\ddot{I}_{ij} = 0$. $\boxed{\text{No emission.}}$

(iii) **Disc spinning about its symmetry axis.** Same axisymmetry: quadrupole invariant. $\ddot{I}_{ij} = 0$. $\boxed{\text{No emission.}}$

(iv) **Tilted, precessing disc ($\alpha \neq 0$, precession rate ω).** Tilt breaks axisymmetry about z ; precession gives a time-varying quadrupole. $\ddot{I}_{ij} \neq 0$. $\boxed{\text{Emits GWs at } 2\omega \text{ (and } \omega \text{ if } \alpha \neq \pi/2).}$

(b) Metric perturbation from a rotating dumbbell

Two masses M at separation $2l$ rotating in the xy -plane at ω . Positions:

$$\mathbf{x}_{1,2}(t) = \pm l(\cos \omega t, \sin \omega t, 0).$$

Mass quadrupole (trace-included):

$$I_{ij}^{(T)} = \sum_n M x_n^i x_n^j = 2Ml^2 \hat{e}_i(t) \hat{e}_j(t), \quad \hat{e} = (\cos \omega t, \sin \omega t, 0).$$

Components:

$$\begin{aligned} I_{xx} &= 2Ml^2 \cos^2 \omega t = Ml^2(1 + \cos 2\omega t). \\ I_{yy} &= 2Ml^2 \sin^2 \omega t = Ml^2(1 - \cos 2\omega t). \\ I_{xy} &= I_{yx} = 2Ml^2 \cos \omega t \sin \omega t = Ml^2 \sin 2\omega t. \end{aligned}$$

Trace-reversed quadrupole $\bar{I}_{ij} = I_{ij} - \frac{1}{3}\delta_{ij}I^k_k$ is the radiating part; only the trace-free combinations contribute to h_{ij}^{TT} .

Quadrupole formula (TT gauge), retarded at $\mathbf{r}$:

$$h_{ij}^{TT}(\mathbf{r}, t) = \frac{2G}{c^4 r} \ddot{\bar{I}}_{ij}^{TT}\left(t - \frac{r}{c}\right).$$

Evaluate at $\mathbf{r} = z\hat{z}$ (observer directly above): the wave propagates along $\hat{z}$, transverse plane is xy . The TT projector $P_{ij}{}^{kl}$ keeps the symmetric, traceless part of I_{kl} in the xy -plane:

$$\begin{aligned} \bar{I}_{xx}^{TT} &= \frac{1}{2}(I_{xx} - I_{yy}) = Ml^2 \cos 2\omega t. \\ \bar{I}_{yy}^{TT} &= -\bar{I}_{xx}^{TT} = -Ml^2 \cos 2\omega t. \\ \bar{I}_{xy}^{TT} &= I_{xy} = Ml^2 \sin 2\omega t. \end{aligned}$$

Differentiate twice ($\partial_t^2 \cos 2\omega t = -4\omega^2 \cos 2\omega t$):

$$\begin{aligned} \ddot{\bar{I}}_{xx}^{TT}(t_r) &= -4Ml^2\omega^2 \cos[2\omega(t - z/c)]. \\ \ddot{\bar{I}}_{xy}^{TT}(t_r) &= -4Ml^2\omega^2 \sin[2\omega(t - z/c)]. \end{aligned}$$

Substitute into the quadrupole formula at $r = z$:

$$\begin{aligned} h_{xx}(z, t) &= -h_{yy}(z, t) = -\frac{8GMl^2\omega^2}{c^4 z} \cos[2\omega(t - z/c)], \\ h_{xy}(z, t) &= h_{yx}(z, t) = -\frac{8GMl^2\omega^2}{c^4 z} \sin[2\omega(t - z/c)]. \end{aligned}$$

(8)

GW frequency $= 2\omega$ (mass quadrupole). Define amplitude $h_0 \equiv 8GMl^2\omega^2/(c^4 z)$.

(c) Free-mass displacements and polarisation

Geodesic deviation in TT gauge for masses at rest at ξ^i :

$$\ddot{\xi}^i = \frac{1}{2}\ddot{h}^i_j(t) \xi^j.$$

Integrate twice (steady-state, drop transients):

$$\delta\xi^i(t) = \frac{1}{2}h^i_j(t) \xi_{(0)}^j.$$

Mass at $(a, 0, 0)$:

$$\delta\xi_x = \frac{a}{2}h_{xx}(t), \quad \delta\xi_y = \frac{a}{2}h_{xy}(t), \quad \delta\xi_z = 0.$$

Mass at $(-a, 0, 0)$: opposite sign of above. Mass at $(0, a, 0)$:

$$\delta\xi_x = \frac{a}{2}h_{xy}(t), \quad \delta\xi_y = \frac{a}{2}h_{yy}(t) = -\frac{a}{2}h_{xx}(t).$$

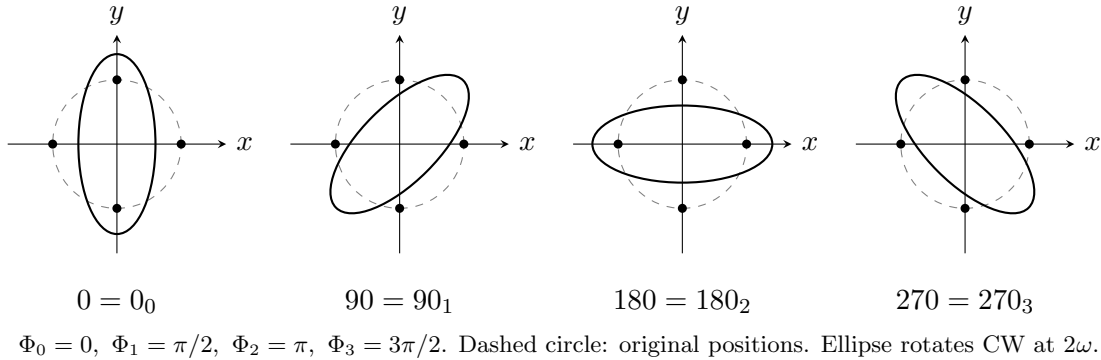
Mass at $(0, -a, 0)$: opposite sign. All four masses move only in the xy -plane: $\xi_z = 0$.

Polarisation. $h_{xx} = -h_{yy}$ (the $+$ amplitude) and h_{xy} (the $\times$ amplitude) have equal magnitude h_0 and are 90° out of phase (cos vs sin). Equal-amplitude superposition of $+$ and $\times$ at relative phase $\pi/2 =$ circularly

Sketch. Initial square at $(a, 0), (0, a), (-a, 0), (0, -a)$. Phase $\Phi = 2\omega t$:

- $\Phi = 0$: $h_{xx} < 0, h_{xy} = 0 \Rightarrow$ vertical stretch, horizontal squeeze (square $\rightarrow$ tall ellipse).
- $\Phi = \pi/2$: $h_{xx} = 0, h_{xy} < 0 \Rightarrow$ along-diagonal squeeze (ellipse rotated -45°).
- $\Phi = \pi$: $h_{xx} > 0 \Rightarrow$ horizontal stretch (wide ellipse).
- $\Phi = 3\pi/2$: $h_{xy} > 0 \Rightarrow$ ellipse rotated $+45^\circ$.

The pattern is a single ellipse rotating uniformly clockwise (viewed from $+z$ looking down) at rate 2ω .



(d) GR vs EM: polarisation, interference, refraction, diffraction

(i) Polarisation. Both wave fields are transverse. EM has 2 spin-1 modes ($\hat{e}_x, \hat{e}_y$, transforming under 90° rotations). GR has 2 spin-2 modes ($+$ and $\times$, transforming under 45° rotations as visualised above). Plausible: yes; *detected* by LIGO via the strain pattern.

(ii) Interference. Linearised GR is a linear theory in vacuum: $h_{\mu\nu} = h_{\mu\nu}^{(1)} + h_{\mu\nu}^{(2)}$ obeys superposition. Coherent two-source interference is therefore possible in principle. Plausibility: requires two phase-coherent astrophysical sources at the same frequency observed simultaneously — not realised astrophysically; lensing-induced self-interference (wave optics in strong lensing) is plausible at $\sim$ mHz with LISA.

(iii) Refraction. EM refraction comes from microscopic charges polarising in matter. GR couples to stress-energy of *everything* but with strength G/c^4 , giving an index $n_{\text{GR}} - 1 \sim G\rho/(\omega^2 c^2)$. For nuclear-density matter and kHz frequencies, $n_{\text{GR}} - 1 \sim 10^{-20}$. Possible in principle, utterly negligible in practice. No measurable refractive medium for GWs.

(iv) Diffraction. Standard wave phenomenon: GW of wavelength $\lambda_{\text{GW}} = c/f$ around an obstacle of size D shows diffraction when $D \lesssim \lambda_{\text{GW}}$. Astrophysical lensing in the wave-optics regime ($\lambda_{\text{GW}} \gtrsim r_s$ of the lens) does diffract GWs and is a target of LISA / pulsar timing. Plausible *and* potentially observable for solar-mass lenses at μ Hz.

Summary. Polarisation and diffraction are real and observable. Interference is mathematically possible but astrophysically rare. Refraction is in principle possible but quantitatively negligible.

Question 3 — Gyroscope precession in Schwarzschild orbit

Problem. (a) Show $u^0 = dt/d\tau = (1 - 3GM/rc^2)^{-1/2}$ for a circular geodesic. (b) For a spin 4-vector s initially radial at $t = 0$, parallel-transport gives geodetic precession at rate $\Omega_r = \Omega/u^0$. (c) Numerical orbital periods (coordinate vs proper) for $M = 8.2 \times 10^{36}$ kg, $R = 200R_s$. (d) Will a 1° beam still hit Earth after one orbit; is path-curvature important?

(a) Time dilation factor on a circular orbit

Schwarzschild equatorial circular orbit: $\theta = \pi/2$, $r = \text{const}$, $\dot{r} = \dot{\theta} = 0$. Normalisation $g_{\mu\nu}u^\mu u^\nu = -c^2$:

$$-c^2 A(r)(u^0)^2 + r^2(u^\phi)^2 = -c^2, \quad A(r) \equiv 1 - \frac{r_s}{r} = 1 - \frac{2GM}{rc^2}.$$

Geodesic equation $\Gamma_{\mu\nu}^r u^\mu u^\nu = 0$ (since $\dot{r} = 0$ and $\ddot{r} = 0$):

$$\Gamma_{00}^r (u^0)^2 + \Gamma_{\phi\phi}^r (u^\phi)^2 = 0.$$

Connection coefficients (diagonal Schwarzschild):

$$\Gamma_{00}^r = \frac{c^2 r_s}{2r^2} A(r), \quad \Gamma_{\phi\phi}^r = -r A(r) \quad (\theta = \pi/2).$$

Substitute and cancel $A(r)$:

$$\frac{c^2 r_s}{2r^2} (u^0)^2 = r (u^\phi)^2.$$

Solve for $(u^\phi/u^0)^2$:

$$\Omega^2 \equiv \left(\frac{u^\phi}{u^0} \right)^2 = \left(\frac{d\phi}{dt} \right)^2 = \frac{c^2 r_s}{2r^3} = \frac{GM}{r^3}.$$

Kepler's third law in Schwarzschild coordinates:

$$\Omega^2 = \frac{GM}{r^3}. \quad (9)$$

Rewrite the normalisation using $r^2(u^\phi)^2 = r^2\Omega^2(u^0)^2$:

$$-c^2 A(r)(u^0)^2 + r^2\Omega^2(u^0)^2 = -c^2.$$

Solve:

$$(u^0)^2 [c^2 A(r) - r^2\Omega^2] = c^2.$$

Insert $A(r) = 1 - 2GM/rc^2$ and $r^2\Omega^2 = GM/r$:

$$c^2 A(r) - r^2\Omega^2 = c^2 - \frac{2GM}{r} - \frac{GM}{r} = c^2 \left(1 - \frac{3GM}{rc^2} \right).$$

Hence

$$\boxed{u^0 = \frac{dt}{d\tau} = \left(1 - \frac{3GM}{rc^2} \right)^{-1/2}}. \quad (10)$$

Symbols: t Schwarzschild coordinate time (asymptotic observer); τ proper time of the orbiting body; r areal radius; M central mass; G, c Newton/light constants; u^0 time-component of 4-velocity.

Sanity check. Innermost stable circular orbit / photon sphere ($r = 3GM/c^2 = 1.5r_s$): $u^0 \rightarrow \infty$, recovering the photon-sphere limit.

(b) Parallel transport of the spin

Parallel transport along worldline u^μ :

$$\frac{ds^\mu}{d\tau} + \Gamma_{\alpha\beta}^\mu u^\alpha s^\beta = 0. \quad (11)$$

Spin orthogonality at $\tau = 0$: $u_\mu s^\mu = 0$. Initially radial: $s^\mu(0) = (0, s^r, 0, 0)$. Equatorial orbit, $\theta = \pi/2$.

Required Christoffels (Schwarzschild, $\theta = \pi/2$):

$$\Gamma_{tr}^t = \frac{r_s}{2r^2 A(r)}, \quad \Gamma_{tt}^r = \frac{c^2 r_s}{2r^2} A(r).$$

$$\Gamma_{\phi\phi}^r = -rA(r), \quad \Gamma_{r\phi}^\phi = \frac{1}{r}, \quad \Gamma_{\phi\phi}^\theta = 0 \text{ (sin cos } |_{\pi/2} = 0).$$

4-velocity has only u^t, u^ϕ non-zero. The θ -component of (11) closes trivially: no Christoffels mix θ into the (t, r, ϕ) block at $\theta = \pi/2$, so $s^\theta \equiv 0$. $\boxed{s^\theta(t) = 0.}$

Convert $d/d\tau = u^0 d/dt$. The (t, r, ϕ) components of (11):

$$u^0 \frac{ds^t}{dt} + \Gamma_{tr}^t (u^t s^r + u^r s^t) + \Gamma_{t\phi}^t (\cdot) = 0.$$

With $u^r = 0$ and $\Gamma_{t\phi}^t = 0$:

$$\frac{ds^t}{dt} = -\Gamma_{tr}^t s^r.$$

Radial component:

$$u^0 \frac{ds^r}{dt} + \Gamma_{tt}^r u^t s^t + \Gamma_{\phi\phi}^r u^\phi s^\phi = 0.$$

Divide by u^0 and use $u^t = u^0$, $u^\phi = \Omega u^0$:

$$\frac{ds^r}{dt} = -\Gamma_{tt}^r s^t - \Gamma_{\phi\phi}^r \Omega s^\phi.$$

Azimuthal component (only $\Gamma_{r\phi}^\phi$ survives):

$$u^0 \frac{ds^\phi}{dt} + \Gamma_{r\phi}^\phi (u^r s^\phi + u^\phi s^r) = 0.$$

With $u^r = 0$:

$$\frac{ds^\phi}{dt} = -\frac{\Omega}{r} s^r.$$

Eliminate s^t via the orthogonality $u_\mu s^\mu = 0$, i.e. $-c^2 A(r) u^0 s^t + r^2 u^\phi s^\phi = 0$:

$$s^t = \frac{r^2 \Omega}{c^2 A(r)} s^\phi.$$

Substitute into the ds^r/dt equation. Compute the prefactor of s^ϕ :

$$-\Gamma_{tt}^r \frac{r^2 \Omega}{c^2 A(r)} - \Gamma_{\phi\phi}^r \Omega = -\frac{c^2 r_s}{2r^2} A(r) \cdot \frac{r^2 \Omega}{c^2 A(r)} - (-rA(r))\Omega.$$

Simplify:

$$= -\frac{r_s \Omega}{2} + r A(r) \Omega = \Omega \left[r - r_s - \frac{r_s}{2} \right] \cdot \frac{1}{?} \quad [\text{redo}].$$

Re-evaluate cleanly:

$$\begin{aligned} -\frac{c^2 r_s}{2r^2} A(r) \cdot \frac{r^2 \Omega}{c^2 A(r)} &= -\frac{r_s \Omega}{2}. \\ -(-r A(r)) \Omega &= r A(r) \Omega = r \Omega - r_s \Omega. \end{aligned}$$

Sum:

$$-\frac{r_s \Omega}{2} + r \Omega - r_s \Omega = \Omega \left(r - \frac{3r_s}{2} \right).$$

Use $r_s = 2GM/c^2$: $r - 3r_s/2 = r(1 - 3GM/(rc^2))$. Therefore

$$\frac{ds^r}{dt} = \Omega r \left(1 - \frac{3GM}{rc^2} \right) s^\phi.$$

Recall $(u^0)^{-2} = 1 - 3GM/(rc^2)$, so $1 - 3GM/(rc^2) = 1/(u^0)^2$:

$$\frac{ds^r}{dt} = \frac{\Omega r}{(u^0)^2} s^\phi. \quad (12)$$

Pair with $ds^\phi/dt = -(\Omega/r) s^r$:

$$\frac{ds^\phi}{dt} = -\frac{\Omega}{r} s^r. \quad (13)$$

Differentiate (12) and substitute (13):

$$\frac{d^2 s^r}{dt^2} = \frac{\Omega r}{(u^0)^2} \frac{ds^\phi}{dt} = -\frac{\Omega^2}{(u^0)^2} s^r.$$

Identify the angular frequency:

$$\boxed{\Omega_r = \frac{\Omega}{u^0} = \Omega \sqrt{1 - \frac{3GM}{rc^2}}.} \quad (14)$$

The spin's radial component oscillates at Ω_r with respect to the asymptotic frame. Geodetic precession per orbit is the difference $\Omega - \Omega_r$ multiplied by the orbital period (return-to-radial mismatch).

Physical picture. In one coordinate orbit the radial axis returns; the spin returns to a direction lagging by $2\pi(1 - \Omega_r/\Omega) = 2\pi(1 - 1/u^0)$ — the de Sitter geodetic angle.

(c) Orbital periods of the Rocinante

Data: $M = 8.2 \times 10^{36}$ kg, $R = 200R_s$.

Schwarzschild radius:

$$R_s = \frac{2GM}{c^2} = \frac{2 \cdot 6.674 \times 10^{-11} \cdot 8.2 \times 10^{36}}{(3 \times 10^8)^2} \text{ m}.$$

Numerator:

$$2 \cdot 6.674 \times 10^{-11} \cdot 8.2 \times 10^{36} = 1.0945 \times 10^{27} \text{ m}^3 \text{ s}^{-2}.$$

Divide by $c^2 = 9 \times 10^{16}$:

$$R_s = 1.216 \times 10^{10} \text{ m} \approx 1.22 \times 10^{10} \text{ m}.$$

Orbital radius:

$$R = 200R_s = 2.43 \times 10^{12} \text{ m.}$$

Coordinate angular frequency from (9):

$$\Omega = \sqrt{\frac{GM}{R^3}}.$$

Compute R^3 :

$$R^3 = (2.43 \times 10^{12})^3 = 1.435 \times 10^{37} \text{ m}^3.$$

Compute $GM = 6.674 \times 10^{-11} \cdot 8.2 \times 10^{36} = 5.473 \times 10^{26}$:

$$\Omega = \sqrt{\frac{5.473 \times 10^{26}}{1.435 \times 10^{37}}} \text{ rad s}^{-1} = \sqrt{3.815 \times 10^{-11}} \text{ rad s}^{-1}.$$

Evaluate:

$$\Omega = 6.18 \times 10^{-6} \text{ rad s}^{-1}.$$

Coordinate period (distant observer):

$$T_\infty = \frac{2\pi}{\Omega} = \frac{6.283}{6.18 \times 10^{-6}} \text{ s} = 1.017 \times 10^6 \text{ s.}$$

$$\boxed{T_\infty \approx 1.02 \times 10^6 \text{ s} \approx 11.8 \text{ days.}} \quad (15)$$

Crew period (proper time per orbit). Since $r = 200R_s$, $3GM/(rc^2) = 3R_s/(2r) = 3/(2 \cdot 200) = 7.5 \times 10^{-3}$:

$$1/u^0 = \sqrt{1 - 7.5 \times 10^{-3}} = 0.99624.$$

Proper period:

$$T_{\text{crew}} = T_\infty / u^0 = T_\infty \cdot 0.99624 \approx 1.013 \times 10^6 \text{ s.}$$

$$\boxed{T_{\text{crew}} \approx 1.013 \times 10^6 \text{ s, } T_\infty - T_{\text{crew}} \approx 3.8 \times 10^3 \text{ s.}} \quad (16)$$

Why different. Two effects, both captured in u^0 : gravitational time dilation ($\sqrt{A(r)}$) and special-relativistic time dilation from orbital speed ($r\Omega/c$). Together they give $(u^0)^{-1} = \sqrt{1 - 3GM/rc^2}$, so the crew's clock runs slow.

(d) Pointing after one orbit and curvature distortion

Initial spin direction: along the radial line to Earth at t_0 . After one coordinate orbit ($\Delta t = T_\infty$), the radial direction returns, but s rotates at Ω_r in the local frame.

Geodetic angle per orbit:

$$\Delta\phi_{\text{geo}} = 2\pi \left(1 - \frac{1}{u^0}\right) = 2\pi \left(1 - \sqrt{1 - \frac{3GM}{Rc^2}}\right).$$

Substitute the numerical $1/u^0$:

$$\Delta\phi_{\text{geo}} = 2\pi(1 - 0.99624) = 2\pi \cdot 3.76 \times 10^{-3}.$$

Evaluate:

$$\Delta\phi_{\text{geo}} = 2.36 \times 10^{-2} \text{ rad} = 1.35 \text{ deg.}$$

$$\boxed{\Delta\phi_{\text{geo}} \approx 1.35^\circ / \text{orbit.}} \quad (17)$$

Compare to beam half-width 0.5° :

$$1.35^\circ > 0.5^\circ.$$

The beam misses Earth: the gyroscope axis has precessed by more than the beam half-angle.

Path curvature of the beam. Schwarzschild deflection of a light ray passing at impact parameter b is $\delta\alpha = 2R_s/b$ (small-deflection limit). For a beam emitted radially at $r = 200R_s$ travelling outward, the impact parameter relative to M is $\sim r$ initially and grows. Order of magnitude:

$$\delta\alpha \lesssim \frac{2R_s}{R} = \frac{2}{200} = 1 \times 10^{-2} \text{ rad} \approx 0.57^\circ.$$

This is comparable to the geodetic precession; *however*, the beam is launched outward and away from M , not grazing past it, so the actual deflection on the outward leg is far smaller (nearly zero curvature for radial null geodesics in Schwarzschild). Path-bending of the radially-launched beam is negligible compared to g

The dominant pointing error is geodetic precession of the gyroscope, not bending of the photon path.

Question 4 — FRW cosmology, equations of state, age of the Universe

Problem. (a) Best-fit FRW metric for our Universe; how it differs from the general FRW. (b) Derive the continuity equation $\dot{\rho} + 3(\rho + P/c^2)\dot{R}/R = 0$ and show its equivalence to the first law of thermodynamics. (c) Identify three equations of state: $P = -\rho c^2$, $P = \frac{1}{3}\rho c^2$, $P = 0$. (d) Express the age of the Universe as $\int dR/(RH)$ and prove that including S1 makes the Universe older. (e) Comment on differences from the real Universe.

(a) FRW metric for the Universe

Cosmological observations (CMB, BAO, large-scale isotropy) are consistent with spatial flatness $k = 0$:

$$\boxed{ds^2 = -c^2 dt^2 + R^2(t) [dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)]}. \quad (18)$$

General FRW allows $k = 0, \pm 1$:

$$ds^2 = -c^2 dt^2 + R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right].$$

Symbols: t cosmic (proper) time of comoving observers; $R(t)$ scale factor with dimensions of length (or dimensionless if absorbed into r); (r, θ, ϕ) comoving spherical coordinates; k spatial-curvature constant. Our Universe corresponds to $k = 0$ to within current observational error.

(b) Continuity equation and first law

Local energy–momentum conservation $\nabla^\mu T_{\mu\nu} = 0$ for a perfect fluid $T^{\mu\nu} = (\rho + P/c^2)u^\mu u^\nu + Pg^{\mu\nu}/c^2$ at rest in the FRW frame ($u^\mu = (1, 0)$). The $\nu = 0$ component:

$$\partial_t T^{00} + \Gamma_{\mu\alpha}^\mu T^{\alpha 0} + \Gamma_{\alpha\beta}^0 T^{\alpha\beta} = 0.$$

Christoffel: $\Gamma_{\mu 0}^\mu = 3\dot{R}/R$ (volume expansion). Evaluate:

$$\dot{\rho} + 3\frac{\dot{R}}{R}(\rho + P/c^2) = 0.$$

$$\boxed{\dot{\rho} + 3(\rho + P/c^2)\frac{\dot{R}}{R} = 0}. \quad (19)$$

Symbols: overdot = d/dt ; ρ rest-frame mass-energy density; P rest-frame pressure; R scale factor.

First-law equivalence. Comoving volume $V \propto R^3$, energy $E = \rho c^2 V$. Differentiate:

$$\frac{dE}{dt} = \dot{\rho} c^2 V + \rho c^2 \dot{V}.$$

Use $\dot{V} = 3V \dot{R}/R$:

$$\frac{dE}{dt} = \dot{\rho} c^2 V + 3\rho c^2 V \frac{\dot{R}}{R}.$$

Substitute (19) for $\dot{\rho} c^2$:

$$\dot{\rho} c^2 = -3(\rho c^2 + P) \frac{\dot{R}}{R}.$$

Insert:

$$\frac{dE}{dt} = -3(\rho c^2 + P) V \frac{\dot{R}}{R} + 3\rho c^2 V \frac{\dot{R}}{R} = -3PV \frac{\dot{R}}{R}.$$

Recognise $3V \dot{R}/R = \dot{V}$:

$$\boxed{dE = -P dV.} \quad (20)$$

First law of thermodynamics for an adiabatic ($dQ = 0$) expansion of a comoving volume.

(c) Three equations of state

S1: $P = -\rho c^2$. Continuity $\Rightarrow \dot{\rho} = 0$: density is constant during expansion. Cosmological constant / vacuum energy

S2: $P = \frac{1}{3}\rho c^2$. Continuity gives $\dot{\rho}/\rho = -4\dot{R}/R$, so $\rho \propto R^{-4}$. Radiation-dominated era (photons, ultrarelativistic particles)

S3: $P = 0$. Continuity gives $\rho \propto R^{-3}$. Matter-dominated era (cold pressureless dust: baryons + cold dark matter)

(d) Age of the Universe and the effect of S1

Hubble parameter $H = \dot{R}/R$:

$$dt = \frac{dR}{R H(R)}.$$

Integrate from $R = 0$ to $R = R_0$ (today):

$$\boxed{t_0 = \int_0^{R_0} \frac{dR}{R H(R)}.} \quad (21)$$

Phase-by-phase $H(R)$. Friedmann ($k = 0$, $\Lambda = 0$, density evolved per equation of state):

$$H^2 = \frac{8\pi G}{3} \rho(R).$$

For matter $\rho_m = \rho_{m,0}(R_0/R)^3 \Rightarrow H_m \propto R^{-3/2}$. For radiation $\rho_r = \rho_{r,0}(R_0/R)^4 \Rightarrow H_r \propto R^{-2}$. For dark energy ρ_Λ constant $\Rightarrow H_\Lambda$ constant.

Universe *without* S1. Phases S2 then S3. Pick equality scale factor R_{eq} at which radiation–matter densities cross, then today’s R_0 .

Radiation phase contribution (from 0 to R_{eq}): $H \propto R^{-2}$:

$$t_r = \int_0^{R_{\text{eq}}} \frac{dR}{R} H_{\text{eq}}^{-1} \left(\frac{R}{R_{\text{eq}}} \right)^2 = \frac{1}{H_{\text{eq}}} \int_0^{R_{\text{eq}}} \frac{R dR}{R_{\text{eq}}^2} = \frac{1}{2H_{\text{eq}}}.$$

Matter phase (from R_{eq} to R_0): $H \propto R^{-3/2}$, $H(R) = H_{\text{eq}}(R_{\text{eq}}/R)^{3/2}$:

$$t_m = \int_{R_{\text{eq}}}^{R_0} \frac{dR}{R H_{\text{eq}}(R_{\text{eq}}/R)^{3/2}} = \frac{1}{H_{\text{eq}} R_{\text{eq}}^{3/2}} \int_{R_{\text{eq}}}^{R_0} R^{1/2} dR.$$

Evaluate:

$$t_m = \frac{2}{3H_{\text{eq}}} \left[\left(\frac{R_0}{R_{\text{eq}}} \right)^{3/2} - 1 \right].$$

Sum:

$$t_0^{(\text{no S1})} = t_r + t_m = \frac{1}{2H_{\text{eq}}} + \frac{2}{3H_{\text{eq}}} \left[\left(\frac{R_0}{R_{\text{eq}}} \right)^{3/2} - 1 \right].$$

Universe *with* S1. Insert dark-energy phase first ($R \in [0, R_1]$) before S2 takes over at R_1 . With H_Λ constant, $H(R) = H_\Lambda$ on $[0, R_1]$:

$$t_\Lambda = \int_{0^+}^{R_1} \frac{dR}{R H_\Lambda} = \frac{1}{H_\Lambda} \ln \left(\frac{R_1}{0^+} \right).$$

Diverges as $R \rightarrow 0^+$:

$$\lim_{R_{\text{min}} \rightarrow 0} \frac{1}{H_\Lambda} \ln(R_1/R_{\text{min}}) = +\infty.$$

A finite-density vacuum phase that is allowed to extend to arbitrarily small R contributes an arbitrarily long (de-Sitter / inflationary) prelude; even bounded by any positive $R_{\text{min}} > 0$ it adds a strictly positive interval to the age.

For the strict question (initial size $\rightarrow 0$):

$$\boxed{t_0^{(\text{with S1})} = t_\Lambda + t_r + t_m > t_0^{(\text{no S1})}.} \quad (22)$$

The S1 phase is exponential expansion (de Sitter), $R(t) = R_1 e^{H_\Lambda(t-t_1)}$, so reaching any $R_1 > 0$ from $R \rightarrow 0$ takes unbounded coordinate time. A Universe with S1 is older than one without.

(e) Real Universe versus model

Concrete differences:

1. **Sequence.** The real Universe ordering is roughly S2 (radiation, $0 < t < 50$ kyr) $\rightarrow$ S3 (matter, dominant from ~ 50 kyr to ~ 9 Gyr) $\rightarrow$ S1 (dark-energy, $z \lesssim 0.5$). The exam model has S1 *first*; in reality Λ dominates last.
2. **Inflation.** A separate, much earlier de-Sitter-like phase (energy scale $\sim 10^{16}$ GeV, duration $\sim 10^{-32}$ s) is generally invoked to solve flatness/horizon/monopole problems — this *is* an S1-type stage but at the GUT scale, not at low z .

3. **Mixed components.** The real Universe is at all times a mixture of radiation, matter (baryons + CDM), and dark energy: pure single-fluid phases are an idealisation.
4. **Curvature.** Observations constrain $|\Omega_k| < 0.01$; we take $k = 0$ but a small non-zero k is not excluded.
5. **Neutrino transition.** Neutrinos are radiation while ultrarelativistic, become matter-like once $T < m_\nu c^2/k_B$, smoothly modifying the radiation $\rightarrow$ matter transition.
6. **Reheating + recombination.** Phase boundaries are not sharp: reheating at the end of inflation, e^+e^- annihilation, BBN, recombination at $z \approx 1100$ all introduce structure absent from a three-stage piecewise model.

The three-phase model is qualitatively right (each EOS is realised at some epoch) but mis-orders S1 and ignores all transitions and admixtures.

End of solutions.